

MEAM 6230 Supplementary Lecture Notes

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Abstract

This document include supplementary lecture notes for the course *MEAM 6230 Learning and Control for Adaptive and Reactive Robots* taught in the University of Pennsylvania by Prof. Figueroa. The lectures are based on foundational topics in the accompanying textbook [1]. These notes serve as extended supplementary material not found in the textbook or sourced from recent research publications. It also serves as a reference to the derivations Prof. Figueroa often writes on the boards in class and extended derivations of functions used in homeworks.

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Lecture 11-12: Re-interpretation of Reference-Based Non-Orthonormal Basis

Note: I wrote part of this derivation on the board during Lecture 11 and I revisit it in Slides 9-11 during Lecture 12.

The main caveat of the normal-based modulation method, where the modulation matrix $M(x)$ for $\dot{x} = M(x)f(x)$ is constructed with the normal-based orthonormal basis $E(x)$ as follows [3],

$$\begin{aligned} E(x) &= \begin{bmatrix} n(x) & e(x) \end{bmatrix}, \\ D(x) &= \text{diag}(\begin{bmatrix} \lambda_1(x) & \lambda_2(x) \end{bmatrix}), \\ M(x) &= E(x)D(x)E(x)^T \end{aligned} \tag{1}$$

where $n(x) = \nabla\Gamma(x)$ is the normal to the boundary of the obstacle surface and $e(x)$ its corresponding tangent. Expanding the normal-based modulation (2) yields the following expression,

$$\begin{aligned} \dot{x} &= E(x)D(x)E(x)^T f(x) \\ &= n(x) \underbrace{\lambda_1 \langle n(x), f(x) \rangle}_{\text{Compress flow in } \hat{n}(x)} + e(x) \underbrace{\lambda_2 \langle e(x), f(x) \rangle}_{\text{Stretch flow in } \hat{e}(x)} \end{aligned} \tag{2}$$

given the scaling eigenvalues defined as,

$$\lambda_1 = 1 - \frac{1}{\Gamma(x)}, \quad \lambda_2 = 1 + \frac{1}{\Gamma(x)} \quad (3)$$

we know that the first term in (2) compressed the nominal flow in the normal direction and stretches it in the tangent direction. Next, I will show that we can derive a similar expansion and interpretation for the terms of the referenced-based modulation approach with modulation matrix $M_r(x)$ [4],

$$\begin{aligned} \dot{x} &= M_r(x)f(x) \\ &= E_r(x)D(x)E_r(x)^{-1}f(x) \end{aligned} \quad (4)$$

and $E_r(x)$ being the non-orthonormal basis for $M_r(x)$ constructed by a reference direction $r(x)$ that creates a linearly independent basis to maintain full-rank of $M_r(x)$. Next, we will show that this condition can be interpreted through the lens of the normal-based orthonormal-basis. Since $r(x)$ cannot be orthogonal to $e(x)$ it is effectively translating $\langle r(x), f(x) \rangle$ onto the basis $e(x)$ axis. Thus, the bases $E(x), E_r(x)$ can be interchanged by stating,

$$\begin{aligned} E_r(x) &= [r(x) \quad e(x)], \quad E(x) = [n(x) \quad e(x)] \\ E_r(x) &= E(x)\Sigma(x) \implies \Sigma(x) = E(x)^T E_r(x) \end{aligned} \quad (5)$$

Then $\Sigma(x) = E(x)^T E_r(x)$ and its inverse take the form of,

$$\begin{aligned} \Sigma(x) &= [n(x) \quad e(x)]^T [r(x) \quad e(x)] & \Sigma(x)^{-1} &= \begin{bmatrix} \frac{1}{n(x)^T r(x)} & 0 \\ -\frac{e(x)^T r(x)}{n(x)^T r(x)} & 1 \end{bmatrix} \\ &= \begin{bmatrix} n(x)^T r(x) & 0 \\ e(x)^T r(x) & 1 \end{bmatrix} \end{aligned} \quad (6)$$

Plugging (5) and (6) into (4) and expanding yields,

$$\begin{aligned} \dot{x} &= M_r(x)f(x) \\ &= [E(x)\Sigma(x)D(x)\Sigma(x)^{-1}E(x)^T] f(x) \\ &= [n(x), e(x)] \begin{bmatrix} \lambda_1(x) & 0 \\ \underbrace{\frac{e(x)^T r(x)(\lambda_1 - \lambda_2)}{n(x)^T r(x)}}_{(3)} & \lambda_2(x) \end{bmatrix} [n(x), e(x)]^T f(x) \\ &= [n(x), e(x)] \underbrace{\begin{bmatrix} \lambda_1(x) & 0 \\ \frac{-2}{\Gamma(x)} \frac{e(x)^T r(x)}{n(x)^T r(x)} & \lambda_2(x) \end{bmatrix}}_{D_r(x)} [n(x), e(x)]^T f(x). \\ &\implies M_r(x) = E(x)D_r(x)E(x)^T \end{aligned} \quad (7)$$

We can see that $D_r(x)$ is a scaling matrix with off-diagonal components corresponding to additional velocities injected in the tangent plane. Thus, expanding (7) yields the following three components:

$$\begin{aligned} \dot{x} &= E(x)D_r(x)E(x)^T f(x) \\ &= n(x) \underbrace{\lambda_1 \langle n(x), f(x) \rangle}_{\text{Compress flow in } \hat{n}(x)} + e(x) \underbrace{\lambda_2 \langle e(x), f(x) \rangle}_{\text{Stretch flow in } \hat{e}(x)} - \underbrace{\frac{2e(x)}{\Gamma(x)} \frac{\langle e(x), r(x) \rangle}{\langle n(x), r(x) \rangle} \langle n(x), f(x) \rangle}_{\text{Injecting additional velocities along } \hat{e}(x)} \end{aligned} \quad (8)$$

Since we used (3) for the eigenvalues, similarly to the normal-based modulation, the first term compresses the nominal flow in the normal direction, the second term stretches the flow in the tangent direction and the third now inject additional velocities in the opposite direction of the second term, effectively pushing the flow away from converging to the spurious attractor on the boundary. Next, I show that this spurious attractor exists when $f(x)$ and $r(x)$ are collinear; i.e., $\frac{f(x)^T r(x)}{\|f(x)\| \cdot \|r(x)\|} = \pm 1$

Let us evaluate (8) at the obstacle boundary; i.e., when $\Gamma(x) = 1 \rightarrow \lambda_1 = 0, \lambda_2 = 2$,

$$\begin{aligned} \dot{x} &= 0 + e(x)2\langle e(x), f(x) \rangle - 2e(x) \frac{\langle e(x), r(x) \rangle}{\langle n(x), r(x) \rangle} \langle n(x), f(x) \rangle \\ &= 2e(x) \left[\langle e(x), f(x) \rangle - \frac{\langle e(x), r(x) \rangle}{\langle n(x), r(x) \rangle} \langle n(x), f(x) \rangle \right] \end{aligned} \quad (9)$$

When $\frac{f(x)^T r(x)}{\|f(x)\| \cdot \|r(x)\|} = \pm 1$ then we can write $f(x) = c \cdot r(x)$ with c being a scalar value, then (9) becomes,

$$\dot{x} = 2ce(x) \left[\langle e(x), r(x) \rangle - \frac{\langle e(x), r(x) \rangle}{\langle n(x), r(x) \rangle} \langle n(x), r(x) \rangle \right] \equiv 0 \quad (10)$$

Thus, assuming that $r(x)$ is not coincident with $n(x)$ or $e(x)$ the only spurious attractor arises from the collinearity issue between $r(x)$ and $f(x)$. Which through the Hairy Ball Theorem, we know that there will be one for convex and star-shaped obstacles as shown in [2].

Lecture 12-13: Derivation of On-Manifold Modulation Approach [2]

Note: Next, I derive the on-manifold modulation approach with variables consistent to the ones used in class and textbook [1], instead of the ones used in the original paper [2], and in 2D for simplicity.

Explicit Spurious Attractor Avoidance

Following the reference-based modulation approach, let us express $D_r(x)$ from (7) as follows,

$$D_r(x) = \begin{bmatrix} \lambda_1(x) & 0 \\ e(x)^T \phi_r(x) & \lambda_2(x) \end{bmatrix} \quad \text{with} \quad \phi_r(x) = \frac{-2}{\Gamma(x)} \frac{r(x)}{\langle n(x), r(x) \rangle} \quad (11)$$

where $\phi_r(x)$ represents the term corresponding to the reference direction scaled by the nominal dynamics that will be projected onto the tangent; i.e., the additional velocities injected on the tangent plane. Next, let us explicitly decompose $D_r(x)$ into diagonal and off-diagonal components,

$$D_r(x) = \underbrace{D(x)}_{(2)} + \begin{bmatrix} 0 \\ e(x)^T \phi_r(x) \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} \quad (12)$$

Thus, a compact expansion of (7) can be written as,

$$\begin{aligned} \dot{x} &= E(x)D_r(x)E(x)^T f(x) \\ &= \underbrace{M(x)f(x)}_{(2)} + e(x)e(x)^T \phi_r(x)n(x)^T f(x) \end{aligned} \quad (13)$$

It would seem that as long as $\phi_r(x) \neq 0$ there will be no spurious attractors induced on the boundary. Nevertheless, because of how $\phi_r(x)$ is defined, as shown in (9), if $f(x)$ and $r(x)$ are collinear; i.e., $\frac{f(x)^T r(x)}{\|f(x)\| \cdot \|r(x)\|} = \pm 1$, the tangential velocities from the normal-based modulation and the off-diagonal

components cancel out. This analysis is one of the insights behind the **on-manifold modulation approach** [2] whose derivation begins by constructing an off-diagonal scaling matrix that **explicitly removes any spurious attractor** due to collinearity of $n(x)$ and $f(x)$ as follows,

$$D_\phi = \begin{bmatrix} \lambda_1(x) & 0 \\ e(x)^T \phi_{\parallel}(x) & \lambda_2(x) \end{bmatrix} \quad \text{with} \quad \phi_{\parallel}(x) = \delta \psi_k(\langle n(x), f(x) \rangle) e(x) \quad (14)$$

where $\phi_{\parallel}(x)$ introduces velocities on the tangent plane and $\psi_k(\langle n(x), f(x) \rangle)$ acts to enable the effect of the additional tangential velocities when $\langle n(x), f(x) \rangle \rightarrow -1$ and disable otherwise with:

$$\psi_k(\langle n(x), f(x) \rangle) = \max\left(0, (-\langle n(x), f(x) \rangle)^k\right) \quad \text{with} \quad k \in [1, 3, 5, \dots] \quad (15)$$

Furthermore, we can alternatively construct $\phi_{\parallel}(x)$ such that its effect is only active at low values of $\Gamma(x)$, i.e., near the obstacle boundary with the following equation:

$$\phi_{\parallel}(x) = \begin{cases} \frac{\delta}{\Gamma(x)^2} \psi_k(\langle n(x), f(x) \rangle) e(x) & \langle e(x), f(x) \rangle \geq 0 \\ 0 & \text{otherwise} \end{cases} \quad (16)$$

Herein, the scaling factor $\frac{\delta}{\Gamma(x)^2}$ ensures that as $\Gamma(x) \rightarrow \infty \implies \phi(x) \rightarrow \mathbf{0} \implies M(x) = \mathbb{I}$. In practice, we set $\delta = 0.1$ to a small number and $k = 1$. Hence, we switch between activating and deactivating the additional tangential velocities governed by the switching function $\langle e(x), f(x) \rangle$ via (??) and (14),

$$\begin{aligned} \dot{x} &= E(x) D_\phi(x) E(x)^T f(x) \\ &= \underbrace{M(x) f(x)}_{(2)} + e(x) e(x)^T \phi(x) \langle n(x), f(x) \rangle \end{aligned} \quad (17)$$

Even though this modulation results in piecewise smooth DS, impenetrability is still preserved and global convergence is ensured for convex obstacles only, see proofs in [2].

γ -manifold Isosurface Tracking for General Obstacle Circumnavigation

While the construction of modulation matrix via (14) allows us to get rid of the nuisance spurious attractors at the boundary, this only works for convex obstacle shapes. To address general obstacle circumnavigation, the **on-manifold approach** in [2] introduces the idea of γ -manifold isosurface tracking. We defined the γ -manifold as the $(d-1)$ -dimensional manifold, $\mathcal{M}_\gamma$, representing the isosurface of $\Gamma(x) = \gamma$ with $\gamma \geq 1$. If $\Gamma(x)$ is a $\mathcal{C}^2$ differentiable scalar field that increases monotonically from the obstacle boundary, then these γ -manifolds are smooth and can be (i) circumnavigated and (ii) used as level-sets to navigate between manifolds. For example, when $\gamma = 1$ the γ -manifold is the exact boundary of the obstacle, when $\gamma > 1$, the γ -manifold corresponds to the shifted-level set of the boundary of the obstacle. We interpret the selected γ -manifold as the *shifted boundary*.

If the robot needs to escape from a concave region one can navigate to a higher *target* manifold, i.e., from $\Gamma(x) = 1 \rightarrow \gamma$ for $\gamma > 1$, to escape from it by performing (i) a **gradient ascent** towards the target γ -manifold when $\Gamma(x) < \gamma$ and (ii) a morphing of tangential **on-manifold dynamics** $\phi(x)$ to circumnavigate the obstacle boundary. Finally, (iii) when $\Gamma(x) > \gamma$ we wish the modulation to perform as the normal-based modulation without **spurious attractors** due to collinearity as in (16). To accomplish this, in [2] we introduced the following composition of 3 scaling matrices:

$$D_{\text{on}}(x) = \underbrace{D_{\nabla}(x, f(x))}_{\text{Gradient Ascent}} + \underbrace{D_{\phi}(x, f(x))}_{\text{Tangent Dynamics}} + \underbrace{\varphi_{\Gamma}(x) D_{\Gamma}(x)}_{\text{Classical Modulation}} \quad (18)$$

which will be used with the orthonormal basis to construct the final on-manifold modulation,

$$\begin{aligned}\dot{x} &= M_{\text{on}}(x)f(x) \\ &= E(x)D_{\text{on}}(x)E(x)^T f(x)\end{aligned}\tag{19}$$

As shown in (18) the first two scaling matrices which modify the dynamics between $1 \leq \Gamma(x) \leq \gamma$ are dependent on the nominal dynamics $f(x)$. This is a clear distinction between the on-manifold method and the reference-based method to create the tangential velocities. Each scaling matrix in (18) provides a desired behavior at different levels of $\Gamma(x)$, while avoiding any spurious attractor:

- For $\Gamma(x) < \gamma$ the **gradient ascent dynamics** will override the dynamics internal to the shifted boundary to follow the path of steepest ascent towards $\Gamma(x) \rightarrow \gamma$.
- When the robot is close and on the shifted boundary $\Gamma(x) = \gamma$ we wish the **tangent space dynamics** $\phi(x)$ to navigate along the obstacle boundary (i.e., circumnavigation).
- When $\Gamma(x) > \gamma$ we seek the behavior to be similar to **classical normal-based modulation**.

Gradient Ascent Dynamics When $\Gamma(x) < \gamma$ navigating to the target boundary can be accomplished via the following scaling matrix:

$$D_{\nabla}(x, f(x)) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \underbrace{[\varphi(n) \quad \varphi'(e)]}_{\Phi(n,e)} \lambda_{11}(x) = \begin{bmatrix} \varphi(n) & \varphi'(e) \\ 0 & 0 \end{bmatrix} \lambda_{11}(x)\tag{20}$$

which will cause the system to track the isosurface at $\Gamma(x) = \gamma$ using both normal and tangent velocity components of the nominal dynamics. λ_{11} , as defined below, is a function that enforces the tracking of the shifted boundary

$$\lambda_{11}(x) = \underbrace{\gamma \left(\frac{1}{\gamma} - \frac{1}{\Gamma(x)} \right)}_{\text{Shifted Boundary}} + \underbrace{\frac{\epsilon \cdot \psi_k(\langle n(x), f(x) \rangle)}{\Gamma(x)^2}}_{\text{Spurious Attractor Avoidance}}\tag{21}$$

with the first component ensuring that the velocity along the normal is > 0 when $\Gamma(x) < \gamma$. The second component ensures that when $\Gamma(x) \geq \gamma$ the morphed dynamics emanated from the obstacle to track the isosurface are never suppressed by adopting (15). Furthermore, the role of the $\Phi(n, e)$ vector function is to flip the direction of the normal and tangent components when the nominal dynamics is moving away from the obstacle. As shown below, $\varphi(n)$ will flip the sign of the normal direction,

$$\varphi(n) = \begin{cases} -1, & \Gamma(x) < \gamma \quad \& \quad n(x)^T f(x) > 0 \\ 1, & \text{otherwise} \end{cases}\tag{22}$$

with the condition $n(x)^T f(x) > 0$ determining whether the flow enters or leaves the obstacle. The same logic is applied for the tangent direction in $\varphi'(e)$,

$$\varphi'(e) = \begin{cases} -1, & \Gamma(x) < \gamma \quad \& \quad e(x)^T f(x) > 0 \\ 1, & \Gamma(x) < \gamma \quad \& \quad e(x)^T f(x) < 0 \\ 0, & \text{otherwise} \end{cases}\tag{23}$$

However, in addition, it is activated only when inside the shifted boundary. By constructing $D_{\nabla}(x, f(x))$ as above we will ensure convergence to the shifted boundary i.e., the γ -manifold with $\gamma > 1$, it negates motion on the tangent space of $\Gamma(x)$ as shown in the modulation expansion below,

$$\begin{aligned}\dot{x} &= E(x)D_{\nabla}(x, f(x))E(x)^T f(x) \\ &= \lambda_{11} \begin{bmatrix} n(x) & e(x) \end{bmatrix} \begin{bmatrix} \varphi(n) & \varphi'(e) \\ 0 & 0 \end{bmatrix} \begin{bmatrix} n(x)^T f(x) \\ e(x)^T f(x) \end{bmatrix} \\ &= \lambda_{11} n(x) [\varphi(n) \langle n(x), f(x) \rangle + \varphi'(e) \langle e(x), f(x) \rangle]\end{aligned}\quad (24)$$

Thus, using $D_{\nabla}(x, f(x))$ only for modulation will bring the system to stop at the target boundary.

Tangent Space Dynamics To alleviate this and deal with non-convex obstacle geometries, we introduce on-manifold tangent dynamics $\phi(x)$ via the following tangent space navigation matrix:

$$D_{\phi}(x, f(x)) = C_{\phi}(x) \underbrace{\text{diag}([0 \quad e(x)^T \phi(x)])}_{\text{Tangential Gating}} \mathbf{1}_n \frac{f(x)^T E(x)}{\|f(x)\|} \quad \text{with} \quad C_{\phi}(x) = \frac{1}{1 + c_{\gamma}(\Gamma(x) - \gamma)^2} \quad (25)$$

This construction results in a rank-1 matrix that defines a local, tangentially constrained, direction-dependent modulation. The tangential gating step expresses $\phi(x)$ on the boundary's orthonormal basis but stripped of its normal component, leaving only tangential steering directions. Then, this tangential strategy is coupled through the outer-product construction, $\frac{f(x)^T E(x)}{\|f(x)\|}$, resulting in a rank-1 correction that redistributes the local flow along the obstacle surface. Finally, C_{ϕ} is a smooth activation function that decays when the far from the shifted boundary γ and strengthens as it gets closer. Geometrically, this matrix reshapes $f(x)$ into a tangential deflection to follow the shifted boundary. c_{γ} is a constant that can be used to control the width of the tracking band; i.e., where the dynamics of $\phi(x)$ are active.

How to construct $\phi(x)$? Here we refer to $\phi(x)$ as the exit strategy that can be used to track desired isosurfaces. In [2] we introduced 3 strategies: (i) Hessian obstacle exit strategy for convex obstacles, (ii) Pseudo-Hessian obstacle exit strategy for simple non-convex obstacles with small concavities and (iii) Geodesic approximation for arbitrary obstacle shapes with highly concave regions. Next, I will describe the Hessian-based approaches with some improved notation for ease of implementation.

Hessian-based obstacle exit strategy for convex obstacles: This approach assumes that an optimal exit point from the obstacle can be defined as $J(x) = \nabla \Gamma(x)^T f(x)$, which can be maximized via:

$$\phi(x) = \underbrace{\nabla^2 \Gamma(x) f(x) + \nabla f(x) \nabla \Gamma(x)}_{\gamma \text{ Isosurface Tracking}} + \underbrace{\delta \cdot \psi_k(\langle n(x), f(x) \rangle) \cdot e(x)}_{\text{Spurious Attractor Avoidance}} \quad (26)$$

where the third term is computed via (15) and $\delta = 0.1$ is a small number. This approach will fail for nonconvex isosurfaces where $J(x)$ has local maxima. It is mostly suitable for convex isosurfaces where $\det(\nabla^2 \Gamma(x)) \neq 0$; i.e., the isosurface has some curvature.

Pseudo-Hessian obstacle exit strategy for nonconvex and planar obstacles: To accommodate for non-convex and planar isosurfaces, one can extend (26) by approximating the target γ -manifold to a convex form. Consider the unit sphere as the approximating manifold then $\Gamma(x) = (x - c_r)^T (x - c_r)$ where c_r is the center of the sphere; i.e., center of the obstacle. For this function, the Hessian is $\nabla^2 \Gamma(x) = 2 \cdot \mathbb{I}$. Thus, given knowledge about the obstacle center, the psuedo-hessian exit strategy is simply:

$$\phi_{\circ}(x) = \underbrace{\nabla^2 \hat{\Gamma}_{\circ}(x) f(x) + \nabla f(x) \nabla \Gamma(x)}_{\gamma \text{ Isosurface Tracking}} + \underbrace{\delta \cdot \psi_k(\langle n(x), f(x) \rangle) \cdot \hat{e}_{\circ}(x)}_{\text{Spurious Attractor Avoidance}} \quad (27)$$

with $\hat{e}_o(x) = \perp \nabla \hat{\Gamma}_o$ and $\nabla^2 \hat{\Gamma}_o(x) = 2 \cdot \mathbb{I}$ being the tangent and Hessian of the approximate spherical manifold centered at c_r ; which is the actual obstacle center. In addition, a small modification should be made to the tangent space navigation matrix to account for this approximation as below:

$$D_{\phi_o}(x, f(x)) = C_{\phi}(x) \underbrace{\text{diag}([0 \quad \hat{e}_o(x)^T \phi_o(x)])}_{\text{Tangential Gating}} \mathbf{1}_n \frac{f(x)^T E(x)}{\|f(x)\|} \quad \text{with} \quad C_{\phi}(x) = \frac{1}{1 + c_{\gamma}(\Gamma(x) - \gamma)^2} \quad (28)$$

This approach works well with nonconvex isosurfaces with regions of small concavity and also for planar isosurfaces. This method is particularly effective when the selected isosurface is further away from the obstacle. Note that, the pseudo-hessian approach (28) can be seen as an generalization of the hessian approach (25) for convex obstacles with radially monotone $\Gamma(x)$ boundary function.

Classical Modulation Following, (18) the third component is the classical modulation from (2) activated only when outside the γ -manifold; i.e., when $\Gamma(x) = \gamma$.

$$D_{\Gamma}(x) = C_{\Gamma}(x) \text{diag}([0 \quad 1]) \left(1 + \frac{1}{\Gamma(x)}\right) \quad \text{with} \quad C_{\Gamma}(x) = \frac{c_{\gamma}(\Gamma(x) - \gamma)^2}{1 + c_{\gamma}(\Gamma(x) - \gamma)^2} \quad (29)$$

the scalar term $C_{\Gamma}(x)$ acts as a blening between the classical modulation and the one induce by $\phi(x)$ from (25) for convex or (28) for non-convex surfaces.

HW4: Analytic Boundary functions $\Gamma(x) : \mathbb{R}^2 \rightarrow \mathbb{R}$ for Star-Shaped Funnel Obstacle

Implicit boundary function for a star-shaped obstacle `gamma_star.m`

Let us define the implicit boundary function for a star-shape obstacle as:

$$\begin{aligned} a &:= x - c_x, & b &:= y - c_y, \\ f(a, b) &= (b^2 - \text{coef}_c)^2 + a^4 \end{aligned} \quad (30)$$

with $\text{coef}_b, \text{coef}_c > 0$ as shape parameters, the boundary function is then constructed as

$$\Gamma(x, y) = f(a, b)^{1/4} - (\text{coef}_c^2 + \text{coef}_b)^{1/4} + 1 \quad (31)$$

By construction, the level set $\Gamma(x, y) = 1$ corresponds to the obstacle boundary

$$f(a, b) = \text{coef}_c^2 + \text{coef}_b. \quad (32)$$

Gradient of $\Gamma(x, y)$: Since $\Gamma(x, y)$ differs from $f(a, b)^{1/4}$ only by an additive constant, its gradient is given by

$$\nabla \Gamma(x, y) = \frac{1}{4} f(a, b)^{-3/4} \nabla f(a, b). \quad (33)$$

Since

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial a} \frac{\partial a}{\partial x} = 4a^3, \quad \frac{\partial f}{\partial y} = \frac{\partial f}{\partial b} \frac{\partial b}{\partial y} = 4b(b^2 - \text{coef}_c). \quad (34)$$

Therefore,

$$\nabla \Gamma(x, y) = \frac{1}{4} f(a, b)^{-3/4} \begin{bmatrix} 4a^3 \\ 4b(b^2 - \text{coef}_c) \end{bmatrix} = \frac{1}{f(a, b)^{3/4}} \begin{bmatrix} a^3 \\ b(b^2 - \text{coef}_c) \end{bmatrix}. \quad (35)$$

Hessian of $\Gamma(x, y)$: Differentiating the gradient once more yields the Hessian matrix

$$\nabla^2 \Gamma(x, y) = \begin{bmatrix} \frac{\partial^2 \Gamma}{\partial x^2} & \frac{\partial^2 \Gamma}{\partial x \partial y} \\ \frac{\partial^2 \Gamma}{\partial y \partial x} & \frac{\partial^2 \Gamma}{\partial y^2} \end{bmatrix}. \quad (36)$$

The first partial derivative wrt. x is

$$\frac{\partial^2 \Gamma}{\partial x^2} = 3a^2 f^{-3/4} + a^3 \left(-\frac{3}{4} \right) f^{-7/4} (4a^3) = \frac{-3a^6 + 3a^2 f(a, b)}{f(a, b)^{7/4}} \quad (37)$$

Similarly, the mixed derivative is

$$\frac{\partial^2 \Gamma}{\partial x \partial y} = \frac{\partial^2 \Gamma}{\partial y \partial x} = a^3 \left(-\frac{3}{4} \right) f^{-7/4} (4b(b^2 - \text{coef}_c)) = \frac{-3a^3 b(b^2 - \text{coef}_c)}{f(a, b)^{7/4}} \quad (38)$$

And the last partial derivative wrt. y is

$$\frac{\partial^2 \Gamma}{\partial y^2} = \frac{-3b^2(b^2 - \text{coef}_c)^2 + (3b^2 - \text{coef}_c)f(a, b)}{f(a, b)^{7/4}} \quad (39)$$

Thus, the Hessian becomes

$$\nabla^2 \Gamma(x, y) = \frac{1}{f(a, b)^{7/4}} \begin{bmatrix} -3a^6 + 3a^2 f(a, b) & -3a^3 b(b^2 - \text{coef}_c) \\ -3a^3 b(b^2 - \text{coef}_c) & -3b^2(b^2 - \text{coef}_c)^2 + (3b^2 - \text{coef}_c)f(a, b) \end{bmatrix} \quad (40)$$

Radially monotone boundary function for a star-shaped obstacle `gamma_star_radial.m`

One can construct a radially monotone boundary function from the implicit boundary function defined in (30). To achieve a boundary function that it monotonically increasing along the radial directions grounded on the obstacle center (c_x, c_y) we define the following variables:

$$\xi = \begin{bmatrix} x \\ y \end{bmatrix}, \quad \xi^c = \begin{bmatrix} c_x \\ c_y \end{bmatrix}, \quad \delta := \xi - \xi^c = \begin{bmatrix} d_x \\ d_y \end{bmatrix} = \begin{bmatrix} x - c_x \\ y - c_y \end{bmatrix}, \quad (41)$$

$$\rho := \|\delta\| = \sqrt{d_x^2 + d_y^2}, \quad \theta := \text{atan2}(d_y, d_x). \quad (42)$$

Then, a radially monotone boundary function from (30) can be defined as,

$$\Gamma(\xi) = 1 + \rho - R(\theta) \quad (43)$$

which satisfies $\Gamma(\xi) = 1$ on the boundary, $\Gamma(\xi) > 1$ outside the obstacle, and $\Gamma(\xi) < 1$ inside the obstacle, where the radial boundary distance $R(\theta)$ has the following closed-form expression

$$R(\theta) = \sqrt{\frac{B(\theta) + \sqrt{B(\theta)^2 + 4A(\theta)b}}{2A(\theta)}}, \quad (44)$$

with

$$A(\theta) = \cos^4 \theta + \sin^4 \theta, \quad B(\theta) = 2c \sin^2 \theta. \quad (45)$$

Moreover, for $\rho > 0$, its **gradient** given by

$$\nabla \Gamma(\xi) = \nabla \rho(\xi) - R'(\theta) \nabla \theta(\xi), \quad (46)$$

and **Hessian** determined as

$$\nabla^2 \Gamma(\xi) = \nabla^2 \rho(\xi) - \left(R''(\theta) \nabla \theta(\xi) \nabla \theta(\xi)^\top + R'(\theta) \nabla^2 \theta(\xi) \right), \quad (47)$$

where

$$\nabla \rho(\xi) = \frac{1}{\rho} \begin{bmatrix} d_x \\ d_y \end{bmatrix}, \quad \nabla \theta(\xi) = \frac{1}{\rho^2} \begin{bmatrix} -d_y \\ d_x \end{bmatrix}, \quad (48)$$

$$\nabla^2 \rho(\xi) = \frac{1}{\rho^3} \begin{bmatrix} d_y^2 & -d_x d_y \\ -d_x d_y & d_x^2 \end{bmatrix}, \quad \nabla^2 \theta(\xi) = \frac{1}{\rho^4} \begin{bmatrix} 2d_x d_y & d_y^2 - d_x^2 \\ d_y^2 - d_x^2 & -2d_x d_y \end{bmatrix}. \quad (49)$$

The first and second derivatives of the radial boundary distance $R(\theta)$ computed as

$$R'(\theta) = \frac{u'(\theta)}{2R(\theta)}, \quad R''(\theta) = \frac{u''(\theta)}{2R(\theta)} - \frac{u'(\theta)^2}{4R(\theta)^3}. \quad (50)$$

with

$$u(\theta) = \frac{B(\theta) + D(\theta)}{2A(\theta)}, \quad R(\theta) = \sqrt{u(\theta)}, \quad Q(\theta) = B(\theta)^2 + 4A(\theta)b, \quad D(\theta) = \sqrt{Q(\theta)}, \quad (51)$$

These auxiliary variables are computed as below:

$$A'(\theta) = -\sin(4\theta), \quad A''(\theta) = -4\cos(4\theta), \quad (52)$$

$$B'(\theta) = 2c \sin(2\theta), \quad B''(\theta) = 4c \cos(2\theta), \quad (53)$$

$$Q'(\theta) = 2B(\theta)B'(\theta) + 4bA'(\theta), \quad Q''(\theta) = 2(B'(\theta)^2 + B(\theta)B''(\theta)) + 4bA''(\theta), \quad (54)$$

$$D'(\theta) = \frac{Q'(\theta)}{2D(\theta)}, \quad D''(\theta) = \frac{Q''(\theta)}{2D(\theta)} - \frac{Q'(\theta)^2}{4D(\theta)^3}, \quad (55)$$

Finally, the derivatives of the auxiliary variable $u(x)$ are defined as,

$$u'(\theta) = \frac{N'(\theta)A(\theta) - N(\theta)A'(\theta)}{2A(\theta)^2}, \quad N(\theta) := B(\theta) + D(\theta), \quad (56)$$

$$u''(\theta) = \frac{M'(\theta)A(\theta) - 2M(\theta)A'(\theta)}{2A(\theta)^3}, \quad M(\theta) := N'(\theta)A(\theta) - N(\theta)A'(\theta). \quad (57)$$

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